Summer Program on Mathematical Modeling 2026

Modeling Spatial Population Growth Using Reaction-Diffusion PDEs

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Contents

Introduction

1. Introduction

Problem Statement

Populations grow in time and spread in space

Many biological populations do not stay in one place while they grow:

- an invasive species colonizes a new region,
- an infectious disease spreads from one area to the next,
- cancer cells invade the surrounding tissue.

Limitation of classical growth models

The exponential and logistic models are ODEs: they describe the **total size** $N(t)$ of a population, but say nothing about **where** the individuals are.

Question

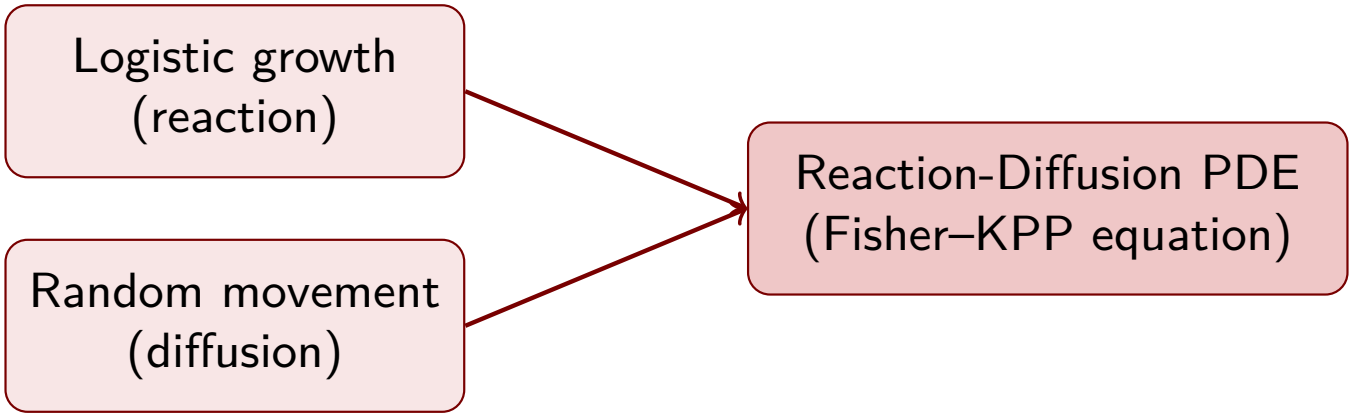
How can we describe the population density N at every position and every time, and see how the population spreads through its habitat?

1. Introduction

Our Project

Idea

Combine **local growth** (logistic model) with **random movement** (diffusion) into a single equation:



Why this project?

Instead of studying ODEs and PDEs only from a theoretical perspective, we apply them to a concrete mathematical-biology problem: from deriving the model, to solving it numerically, to visualizing how the population spreads.

Research Objectives

2. Research Objectives

Goal

Simulate how a population **grows** and **spreads** in space.

1. Model

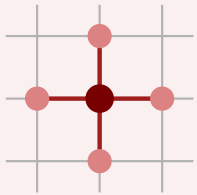
$N(t)$

↓

$N(x, y, t)$

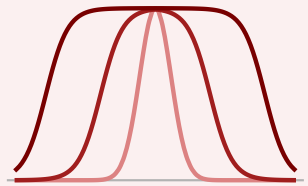
Logistic ODE → Fisher–KPP PDE

2. Discretize



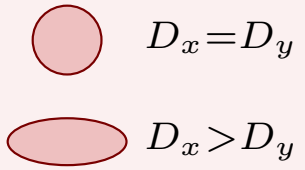
Numerical Methods (FDM, FEM)

3. Simulate



Python code, 1D and 2D

4. Analyze



Effect of D_x, D_y on the spread

Literature Review

3. Literature Review

Historical Background

The Fisher–KPP equation

- **1937:** Fisher [2] proposed the equation to describe the spread of an advantageous gene through a population. In the same year, Kolmogorov, Petrovskii and Piskunov [3] studied it rigorously.
- Main result: the equation has **traveling wave** solutions, i.e. a front that moves at the constant speed

$$c^* = 2\sqrt{rD}.$$

- **1951:** Skellam [5] first applied the model to a real population: muskrats spreading across Central Europe after 1905. The square root of the occupied area grew linearly in time, as the model predicts.

3. Literature Review

Applications in Practice

Field	Phenomenon	Ref.
Ecology	Invasive species (muskrats, butterflies)	[5], [6]
Epidemiology	Rabies spreading among foxes in Europe	[7]
Medicine	Brain tumor (glioma) growth on MRI	[8]
Archaeology	Spread of farming in Neolithic Europe (≈ 1 km/year)	[9]
Experiment	Invasion fronts of microorganisms in the lab	[10]

Common approach

Estimate D and r from data $\Rightarrow$ predict the spreading speed $c^* = 2\sqrt{rD}$ $\Rightarrow$ compare it with the observed spread.

3. Literature Review

Mathematical Modeling: When ODE and PDE Combine

Exponential Population Growth Equation

Firstly, we start with a simple Exponential Population Growth Equation as below:

$$\frac{dN}{dt} = rN$$

Where $N(t)$ is the population size at time t, and r is the growth rate (can be a constant r or a function over time $r(t)$; to be simple we assume that $r = const$).

3. Literature Review

Mathematical Modeling: When ODE and PDE Combine

Logistic Population Growth Equation

Now that with the Carrying Capacity K, we can use a Logistic Model:

$$\frac{dN}{dt} = r\left(1 - \frac{N}{K}\right)N$$

- $N \ll K$: The population grows exponentially.
- $N \to K$: The population's growth speed decreases.
- $N = K$: The population stops growing since $\frac{dN}{dt} = r\left(1 - \frac{N}{N}\right)N = 0.$

3. Literature Review

Mathematical Modeling: When ODE and PDE Combine

Reaction-Diffusion Equation

Supposing that the population starts at the center of a fixed segment on the Ox axis and it gradually spreads. The population density at the point x at time t is described by $N = N(x,t)$:

$$\frac{\partial N}{\partial t} = \underbrace{D\frac{\partial^2 N}{\partial x^2}}_{\text{Diffusion}} + \underbrace{r\Big(1-\frac{N}{K}\Big)N}_{\text{Population Growth}}$$

By normalizing the population density as $u = N/K$, the equation becomes the *Fisher–KPP equation*.

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} + ru\Big(1-u\Big)$$

Where $D > 0$ is the diffusion coefficient, which represents the rate at which the population spreads through space.

3. Literature Review

Mathematical Modeling: Two-Dimensional Space

Reaction-Diffusion Equation on Two-Dimensional Space

Suppose that a population is initially concentrated at the center of a rectangular domain in the Oxy plane and gradually spreads in both the horizontal and vertical directions. The population density at position (x, y) and time t is described by $N = N(x, y, t)$:

$$\frac{\partial N}{\partial t} = D_x \frac{\partial^2 N}{\partial x^2} + D_y \frac{\partial^2 N}{\partial y^2} + r\left(1 - \frac{N}{K}\right)N$$

Where $D_x, D_y > 0$ are the diffusion coefficients representing the rates at which the population spreads in the horizontal and vertical directions, respectively.

3. Literature Review

Mathematical Modeling: Two-Dimensional Space

Reaction-Diffusion Equation on Two-Dimensional Space

If $D_x = D_y$ then we can write in short as:

$$\frac{\partial N}{\partial t} = D\left(\frac{\partial^2 N}{\partial x^2} + \frac{\partial^2 N}{\partial y^2}\right) + r\left(1 - \frac{N}{K}\right)N$$

Or even shorter:

$$\frac{\partial N}{\partial t} = D\nabla^2 N + r\left(1 - \frac{N}{K}\right)N$$

Note that D_x, D_y, and D may be functions of space and time, such as $D_x(x, y, t)$, $D_y(x, y, t)$, and $D(x, y, t)$. However, for simplicity, we assume that they are constants.

3. Literature Review

Mathematical Modeling: Neumann Boundary Condition

No-Flux Boundary Condition

We assume that the population cannot leave the domain. Therefore, there is no population flux across the boundary:

$$\frac{\partial N}{\partial n} = 0 \qquad \text{on } \partial\Omega$$

where $\partial\Omega$ denotes the boundary of the spatial domain.

For a Rectangular Domain

For $\Omega = [0, L_x] \times [0, L_y]$, the Neumann Boundary Condition is:

$$\frac{\partial N}{\partial x}(0, y, t) = \frac{\partial N}{\partial x}(L_x, y, t) = 0,$$

$$\frac{\partial N}{\partial y}(x, 0, t) = \frac{\partial N}{\partial y}(x, L_y, t) = 0.$$

3. Literature Review

Mathematical Modeling: Dirichlet Boundary Condition

Dirichlet Boundary Condition

Instead of specifying the population flux, we specify the population density directly on the boundary:

$$N(x, y, t) = g(x, y, t) \qquad \text{on } \partial\Omega.$$

For example, a zero-density boundary condition is:

$$N(x, y, t) = 0 \qquad \text{on } \partial\Omega.$$

Interpretation

The population density is prescribed at the boundary.
Unlike the Neumann condition, individuals may cross the boundary.
To be simple we use Neumann Boundary Condition in this presentation.

Research Methodology

4. Research Methodology

Spatial Discretization

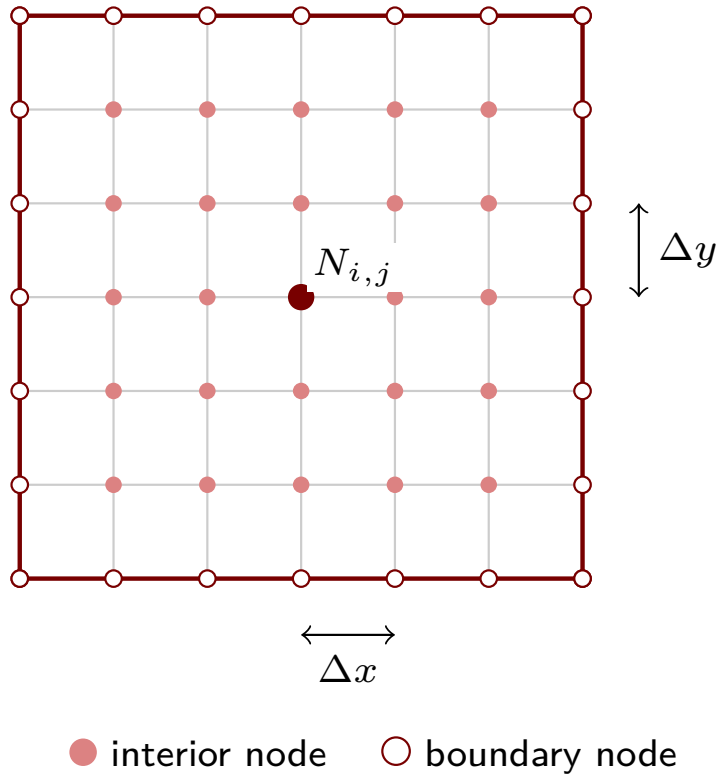
Finite Difference Method

We discretize the spatial domain into a uniform grid:

$$x_i = i\Delta x, \qquad y_j = j\Delta y$$

and approximate the population density by

$$N(x_i, y_j, t) \approx N_{i,j}(t).$$



4. Research Methodology

Second-Order Spatial Derivatives

Second-Order Spatial Derivatives

Using the central difference approximation

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

Then:

$$N_{xx} = \frac{\partial^2 N}{\partial x^2} \approx \frac{N_{i+1,j} - 2N_{i,j} + N_{i-1,j}}{(\Delta x)^2}$$

$$N_{yy} = \frac{\partial^2 N}{\partial y^2} \approx \frac{N_{i,j+1} - 2N_{i,j} + N_{i,j-1}}{(\Delta y)^2}.$$

4. Research Methodology

Time Discretization

Let

$$t_n = n\Delta t, \qquad N_{i,j}^n \approx N(x_i, y_j, t_n).$$

Using the Explicit Euler method:

$$\frac{\partial N}{\partial t} \approx \frac{N_{i,j}^{n+1} - N_{i,j}^n}{\Delta t}.$$

Therefore,

$$N_{i,j}^{n+1} = N_{i,j}^n + \Delta t \Big[D_x N_{xx} + D_y N_{yy} + r N_{i,j}^n \Big(1 - \frac{N_{i,j}^n}{K}\Big)\Big].$$

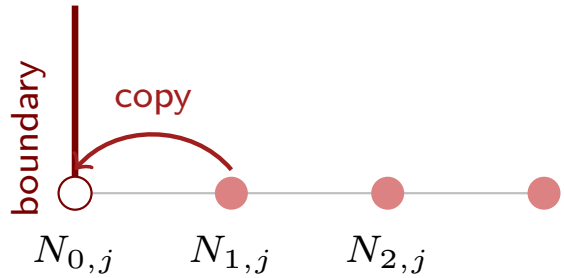
4. Research Methodology

Discrete Boundary Condition

Neumann condition on the grid

At the left edge, replace $\dfrac{\partial N}{\partial x} = 0$ by a one-sided difference:

$$\frac{N_{1,j} - N_{0,j}}{\Delta x} = 0 \implies N_{0,j} = N_{1,j}.$$



After every time step, each boundary node copies its inner neighbor

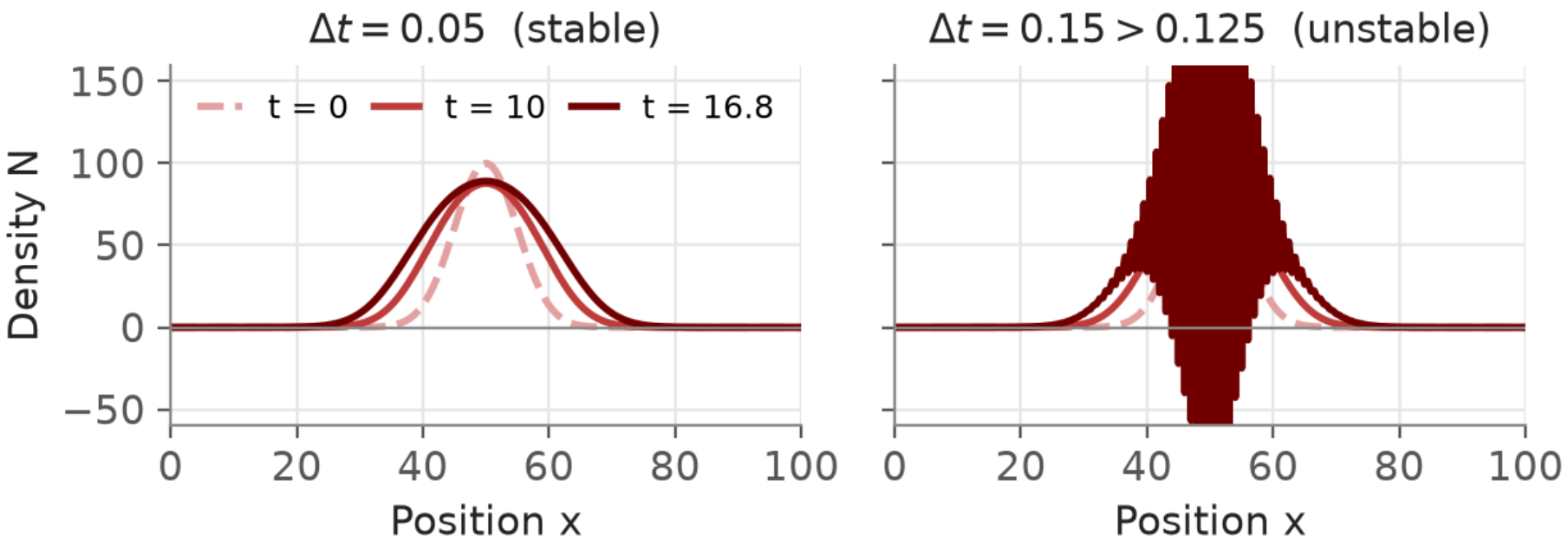
$$N_{0,j} = N_{1,j}, \qquad N_{N_x,j} = N_{N_x-1,j}, \qquad N_{i,0} = N_{i,1}, \qquad N_{i,N_y} = N_{i,N_y-1}.$$

4. Research Methodology

Stability Condition

Courant - Friedrichs - Lewy (CFL) Stability Condition [4]

$$\Delta t \leq \Delta t_{\max} = \frac{1}{2\left(\frac{D_x}{\Delta x^2} + \frac{D_y}{\Delta y^2}\right)} \qquad \left(\text{1D: } \Delta t \leq \frac{\Delta x^2}{2D}\right)$$



4. Research Methodology

Finite Element Method: Weak Formulation

Weak form [11]

Multiply the PDE by a test function v, integrate over Ω and integrate by parts ($f(N) = rN(1 - N/K)$):

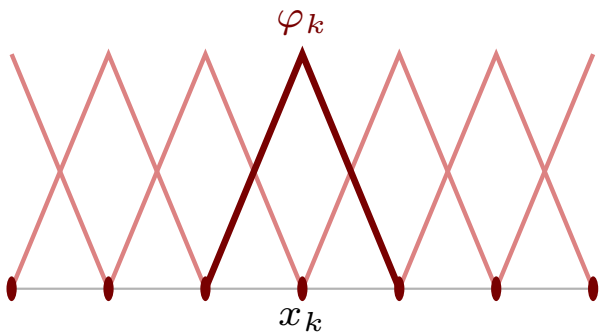
$$\int_\Omega N_t\, v\, d\mathbf{x} + \int_\Omega \left(D_x N_x v_x + D_y N_y v_y\right) d\mathbf{x} = \int_\Omega f(N)\, v\, d\mathbf{x} + \underbrace{\oint_{\partial\Omega} v\left(D_x N_x n_x + D_y N_y n_y\right) ds}_{=0 \text{ by the no-flux condition}}$$

Natural boundary condition

The Neumann condition removes the boundary term: nothing has to be imposed at the boundary.

4. Research Methodology

Finite Element Method: Galerkin Discretization



$\varphi_k = 1$ at x_k, 0 at other nodes

Approximation

$$N(\mathbf{x}, t) \approx \sum_k N_k(t)\, \varphi_k(\mathbf{x})$$

In 2D: bilinear elements $\varphi_i(x)\, \psi_j(y)$ on the same grid as the FDM.

Choosing $v = \varphi_l$ gives a system of ODEs

$$M \frac{d\mathbf{N}}{dt} + S\,\mathbf{N} = M f(\mathbf{N})$$

$$\underbrace{M_{kl} = \int_\Omega \varphi_k \varphi_l,}_{\text{mass matrix}} \qquad \underbrace{S_{kl} = \int_\Omega \left(D_x\, \partial_x \varphi_k\, \partial_x \varphi_l + D_y\, \partial_y \varphi_k\, \partial_y \varphi_l\right)}_{\text{stiffness matrix}}$$

4. Research Methodology

Finite Element Method: Time Stepping

Semi-implicit Euler: diffusion implicit, reaction explicit

$$(M + \Delta t\, S)\, \mathbf{N}^{n+1} = M\big(\mathbf{N}^n + \Delta t\, f(\mathbf{N}^n)\big)$$

No stability limit from diffusion $\Rightarrow$ we can use $\Delta t = 0.5$.

	FDM	**FEM**
Unknowns	values $N_{i,j}$ at grid points	coefficients of φ_k
Neumann condition	copy the inner neighbor	natural (automatic)
Time step	explicit, $\Delta t \leq \Delta t_{\max}$	semi-implicit, larger Δt
Cost per step	cheap update	one linear solve
Domain	rectangular grid	any shape (triangular mesh)

4. Research Methodology

ADI Method: Alternating Directions

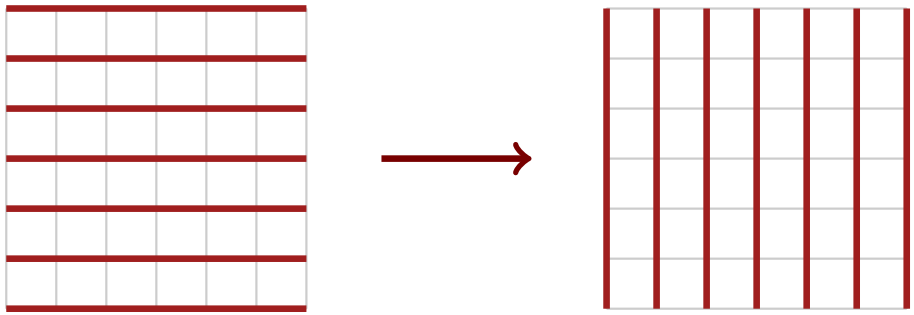
Peaceman–Rachford ADI [4], [12]

Each step is split into two half steps, each implicit in **one** direction only:

$$\left(I - \tfrac{\Delta t}{2} D_x L_x\right) N^* = \left(I + \tfrac{\Delta t}{2} D_y L_y\right) N^n \quad \text{implicit in } x$$

$$\left(I - \tfrac{\Delta t}{2} D_y L_y\right) N^{n+1} = \left(I + \tfrac{\Delta t}{2} D_x L_x\right) N^* \quad \text{implicit in } y$$

L_x, L_y: central second differences; Neumann condition through a ghost point $N_{-1,j} = N_{1,j}$.



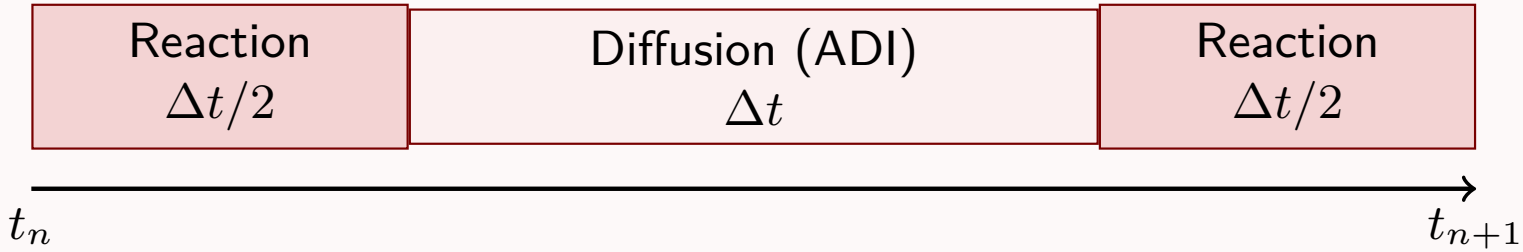
half step 1: one tridiagonal system per row

half step 2: one tridiagonal system per column

4. Research Methodology

ADI Method: Treating the Logistic Term

Strang splitting of one time step [13]



The reaction part has an exact solution (logistic ODE)

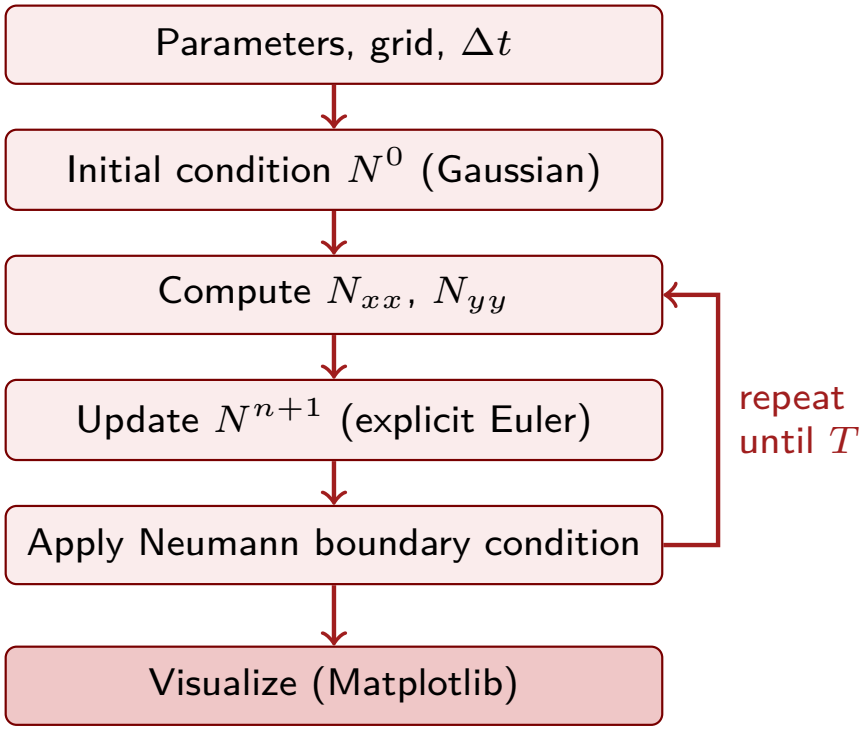
$$N(t) = \frac{K N_0 \, e^{rt}}{K + N_0 \left(e^{rt} - 1\right)}$$

Result

Unconditionally stable and second-order accurate in time: Δt is limited by accuracy only.

4. Research Methodology

Implementation



FEM: loop body = one linear solve

Numerical settings

	1D	**2D**
$\Delta x\ (= \Delta y)$	0.5	1
$\Delta t_{\max}$ (FDM)	0.125	0.1
Δt FDM	0.05	0.08
Δt FEM	0.5	0.5

Tools

Python, NumPy (vectorized arrays), Matplotlib

Results and Discussion

5. Results and Discussion

One-Dimensional Simulation

Suppose we have a population of animals in a closed one-dimensional habitat

$$\Omega = [0, 100]$$

and no animal can cross the boundary of the domain. Therefore, the Neumann Boundary Condition is:

$$\frac{\partial N}{\partial x}(0, t) = \frac{\partial N}{\partial x}(100, t) = 0.$$

We choose the simulation constants:

$$D = 1, \qquad r = 0.1, \qquad K = 100.$$

5. Results and Discussion

One-Dimensional Simulation

We choose the initial population distribution:

$$N(x,0) = N_0 \exp\left[-\frac{(x-x_0)^2}{2\sigma^2}\right].$$

Suppose that the population initially concentrates at the center of the domain, where

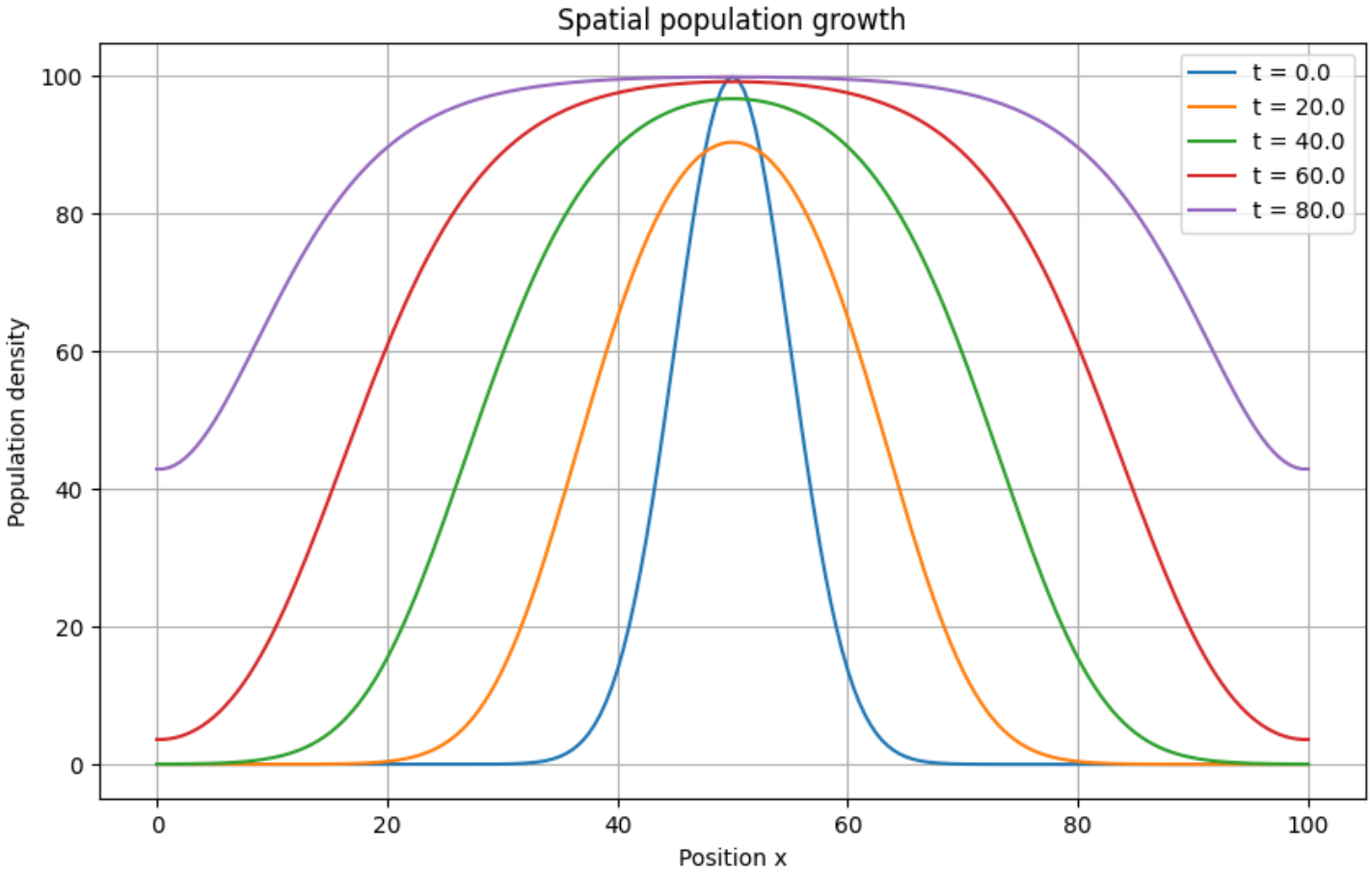
$$x_0 = 50, \qquad N_0 = 100, \qquad \sigma = 5.$$

Then the initial population distribution is:

$$N(x,0) = 100 \exp\left[-\frac{(x-50)^2}{50}\right].$$

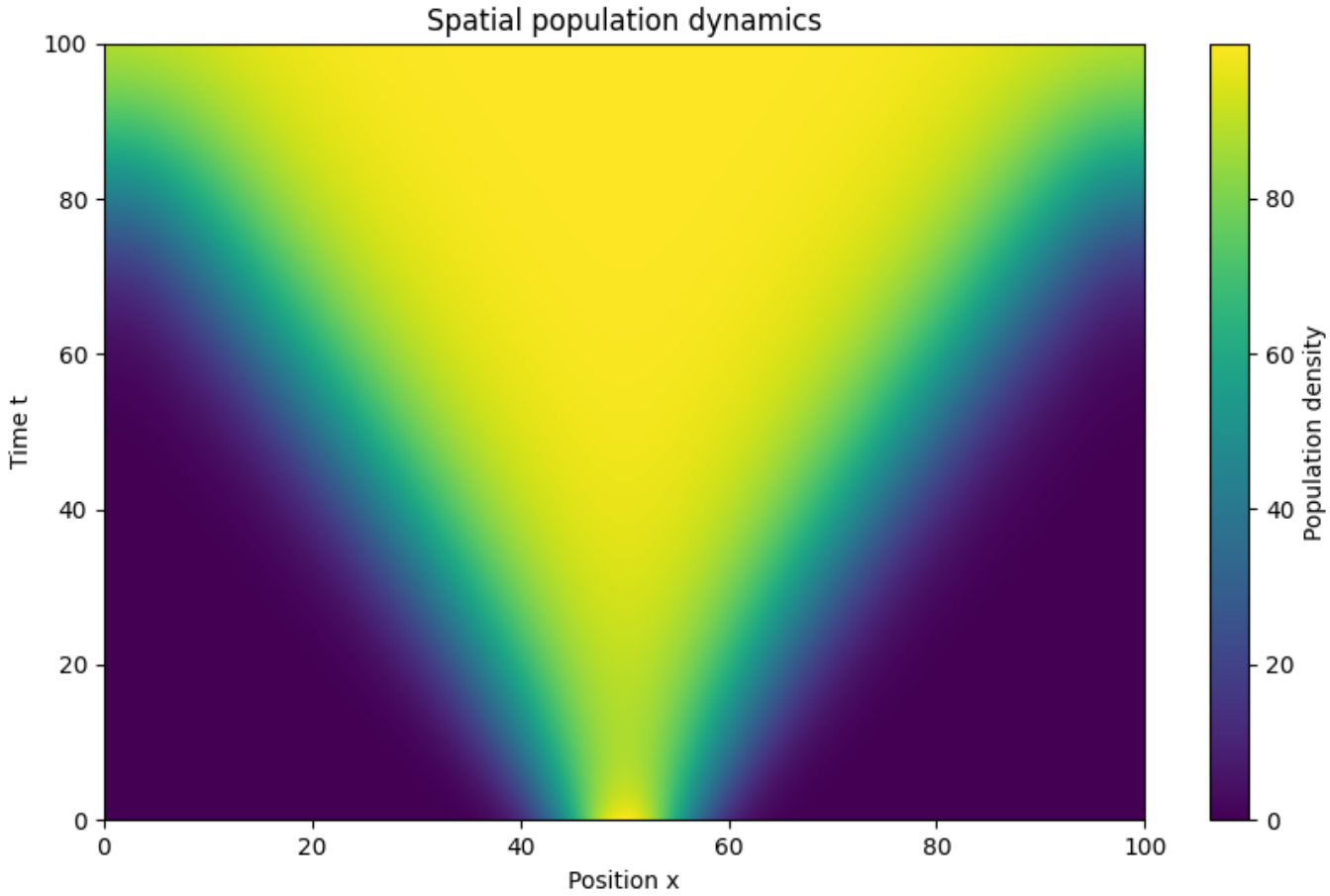
5. Results and Discussion

One-Dimensional Simulation



5. Results and Discussion

One-Dimensional Simulation



5. Results and Discussion

Two-Dimensional Simulation

Suppose we have a population of animals in a closed rectangular domain

$$\Omega = [0, 100] \times [0, 100]$$

and no animals can leave the domain. So the Neumann Boundary Condition is:

$$\frac{\partial N}{\partial x}(0, y, t) = \frac{\partial N}{\partial x}(100, y, t) = 0,$$
$$\frac{\partial N}{\partial y}(x, 0, t) = \frac{\partial N}{\partial y}(x, 100, t) = 0.$$

We choose simulation parameters:

$$D_x = 4, \qquad D_y = 1, \qquad r = 0.1, \qquad K = 100.$$

5. Results and Discussion

Two-Dimensional Simulation

And we choose the initial state:

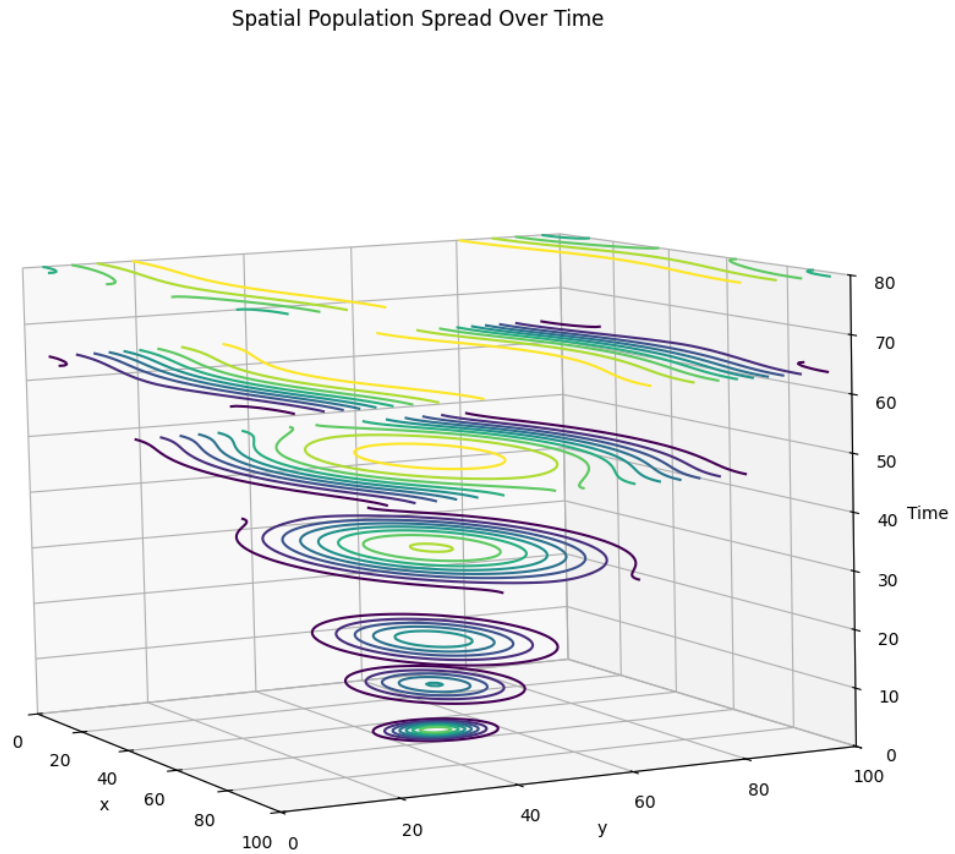
$$N(x, y, 0) = N_0 \exp\Big[- \frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2} \Big]$$

Suppose that the initial location is at the center of the rectangular domain $(x_0, y_0) = (50, 50)$, the maximum initial population density is $N_0 = 80$ and the initial population spread is $\sigma = 5$. Then we have:

$$N(x, y, 0) = 80 \exp\Big[- \frac{(x - 50)^2 + (y - 50)^2}{50} \Big]$$

5. Results and Discussion

Two-Dimensional Simulation



5. Results and Discussion

Real Terrain (Ba Vì): Model and Terrain Functions

Reaction–diffusion–taxis model

$$\frac{\partial N}{\partial t} = \underbrace{\nabla \cdot \big(D\,\nabla N\big)}_{\text{movement}} - \underbrace{\nabla \cdot \big(\chi\, N\, \nabla A\big)}_{\text{drift to good habitat}} + \underbrace{r\, N\Big(1 - \frac{N}{K}\Big)}_{\text{local growth}}$$

D, χ, A, r, K depend on the position $\mathbf{x} = (x, y)$; no-flux boundary; x, y in km, t in years.

Terrain functions, built from the elevation $z(x, y)$

Slope	$s = \lvert\nabla z\rvert$	m per m
Mobility	$D = D_0\, e^{-s/s_0}$ (rivers: $\rho\, D_0$)	slow on slopes
Carrying capacity	$K = K_0\, e^{-(z/z_c)^2}$	poor high up
Growth rate	$r = r_0$ (rivers: r_{riv})	risky crossing
Attraction	$A = K/K_0$	good habitat
Taxis	$\chi = \chi_0\, D/D_0$	drift

5. Results and Discussion

Real Terrain (Ba Vì): Data, Grid and Initial Condition

Data and grid

Elevation	Copernicus DEM GLO-30 (30 m)
Rivers	OpenStreetMap
Region	20.95–21.20°N, 105.20–105.55°E $(36 \times 28$ km$)$
Grid	180×252 cells of 144×154 m (mean of 5×5 DEM pixels)
Time stepping	finite volumes + explicit Euler, $\Delta t = 0.0043$ year, 6918 steps to $T = 30$ years

Initial condition

Start $\mathbf{x}_0$	21.10°N, 105.40°E: plain north-east of Ba Vì, $z = 43$ m
Density	$N(\mathbf{x}, 0) = 80 \exp\big(-|\mathbf{x} - \mathbf{x}_0|^2/2\sigma^2\big)$, $\sigma = 0.5$ km
Arrival	a cell counts as reached when $N \geq 0.2\, K_0 = 20$

5. Results and Discussion

Real Terrain (Ba Vì): Constants

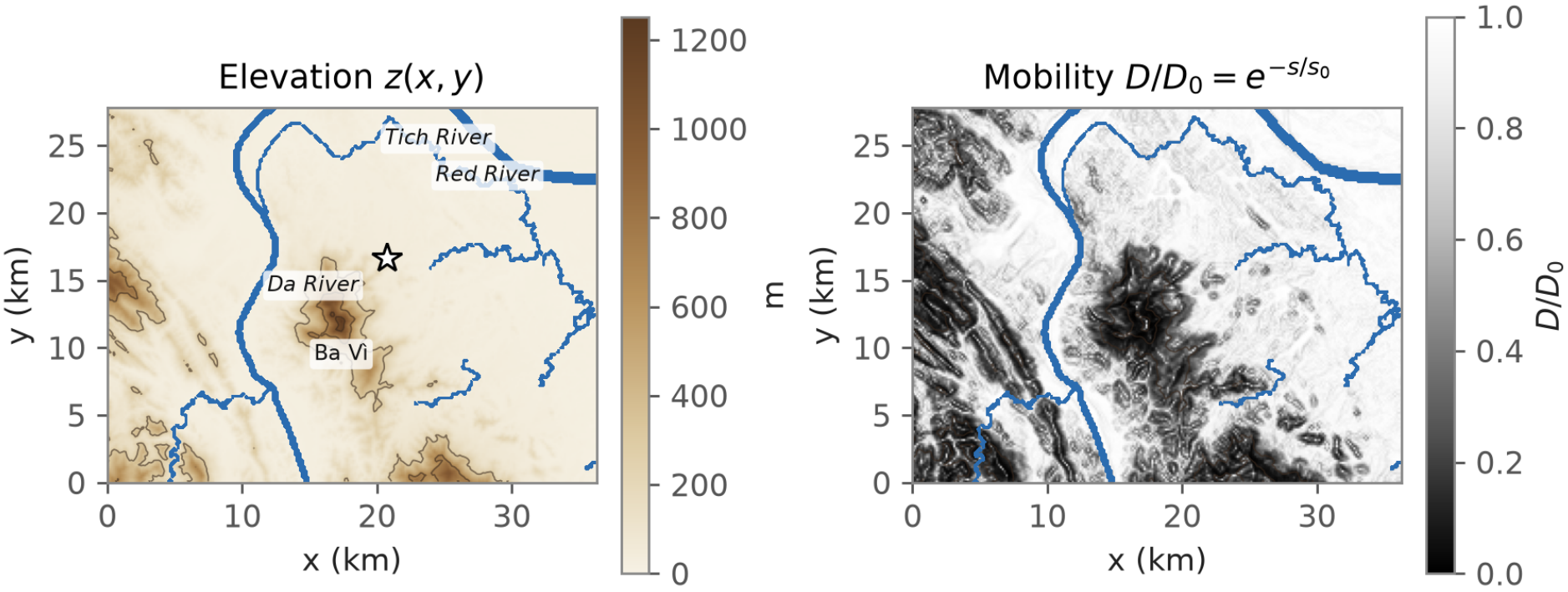
Constant	Value	Meaning
D_0	1 km^2/year	mobility on flat land
r_0	0.5 /year	growth rate on land
K_0	100	carrying capacity in the lowland
χ_0	1 km^2/year	strength of the drift
s_0	0.2	slope scale (slope 0.6: $D \approx 0.05\, D_0$)
z_c	700 m	altitude scale (summit: $K \approx 4$)
$c = 2\sqrt{r_0 D_0}$	1.41 km/year	Fisher–KPP front speed on flat land

River	Width on the grid	$\rho = D/D_0$	r_{riv}
Red River	5 cells ≈ 700 m	0.002	-2 /year
Da River	3 cells ≈ 450 m	0.002	-2 /year
Tich River	1 cell ≈ 150 m	0.1	0
other rivers	1 cell	0.3	0

Values are illustrative, not fitted to a particular species.

5. Results and Discussion

Real Terrain (Ba Vì): Elevation and Mobility

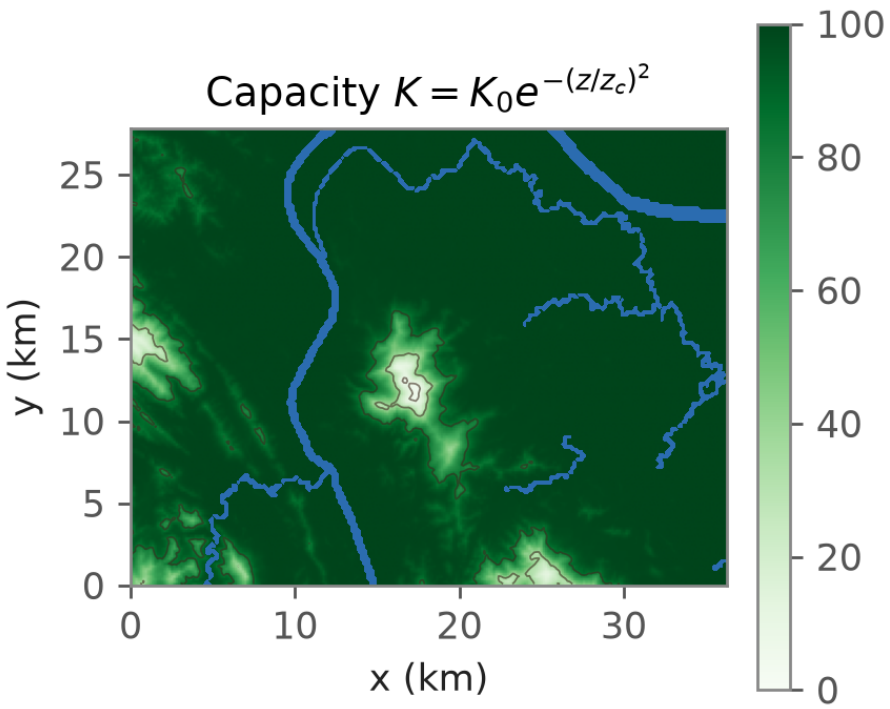


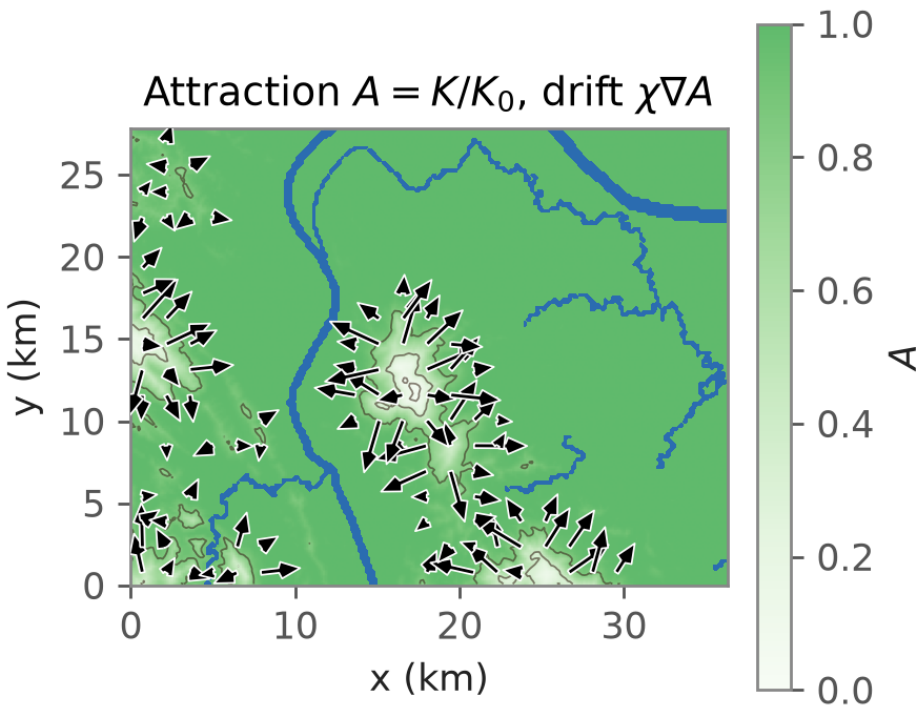
Mobility $D = D_0\, e^{-s/s_0}, \quad s = |\nabla z|$

Flat land: $D \approx D_0$. Steep flanks of Ba Vì: $D < 0.1\, D_0$. Rivers: $D = \rho\, D_0$ (blue).

5. Results and Discussion

Real Terrain (Ba Vì): Habitat and Drift



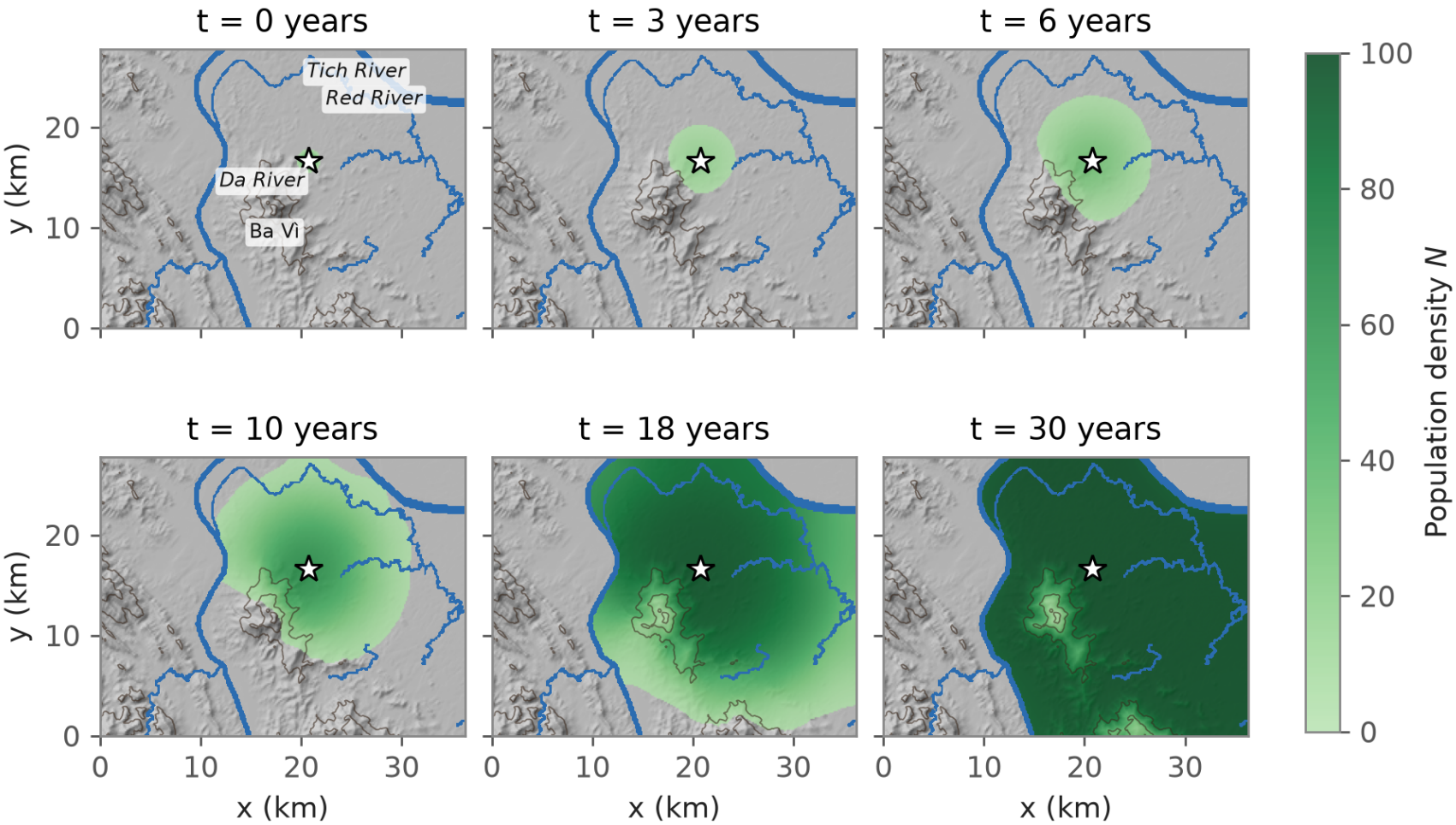


Carrying capacity $K = K_0\, e^{-(z/z_c)^2}$

Lowland: $K \approx K_0 = 100$. Summit: $K \approx 4$. Arrows: drift $\chi\nabla A$ points downhill, toward better habitat.

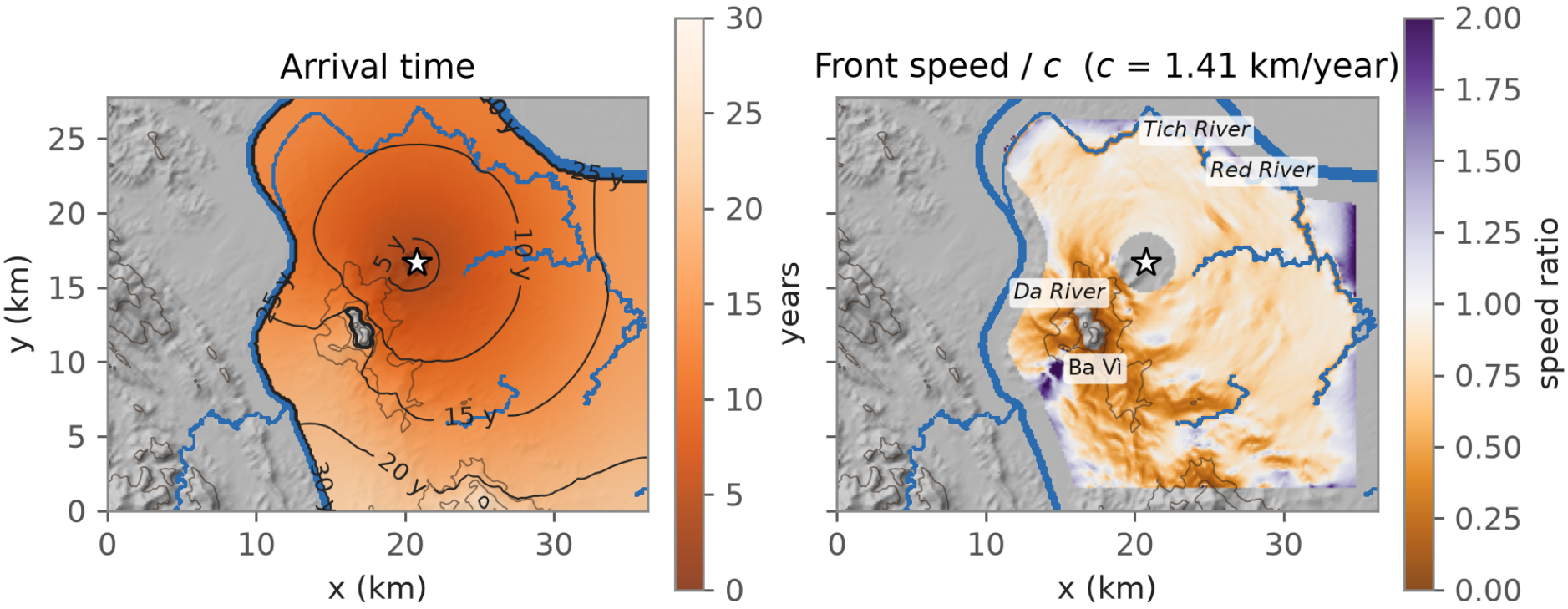
5. Results and Discussion

Real Terrain (Ba Vì): Population over 30 Years



5. Results and Discussion

Real Terrain (Ba Vì): Arrival Time and Front Speed



Front speed $= 1/|\nabla T_{\text{arrival}}|$

Orange: the front slows down. Purple spots: two fronts meet (not a real acceleration). Not reached after 30 years: 36% of the land.

5. Results and Discussion

Real Terrain (Ba Vì): Results

Effect	**Where**	**Result**
Normal spreading	flat plains ($s < 0.05$)	front speed $\approx 0.88\,c$ (median)
Slowed down	steep slopes ($s > 0.3$)	front speed $\approx 0.35\,c$ (median)
Slightly slowed	Tich River	crossed
Blocked	Da River, Red River	far banks empty after 30 years
Not colonized	Ba Vì summit	$K \approx 4 < 20$

Result

After 30 years the population occupies 64% of the land: the far banks of the big rivers and the summit stay empty.

5. Results and Discussion

Case Study: Spread of Farming into Europe

Published study [14]

R. Pinhasi, J. Fort, A. J. Ammerman, "Tracing the Origin and Spread of Agriculture in Europe," *PLoS Biology* 3(12): e410, 2005 (open access, data in Table S1).

Question

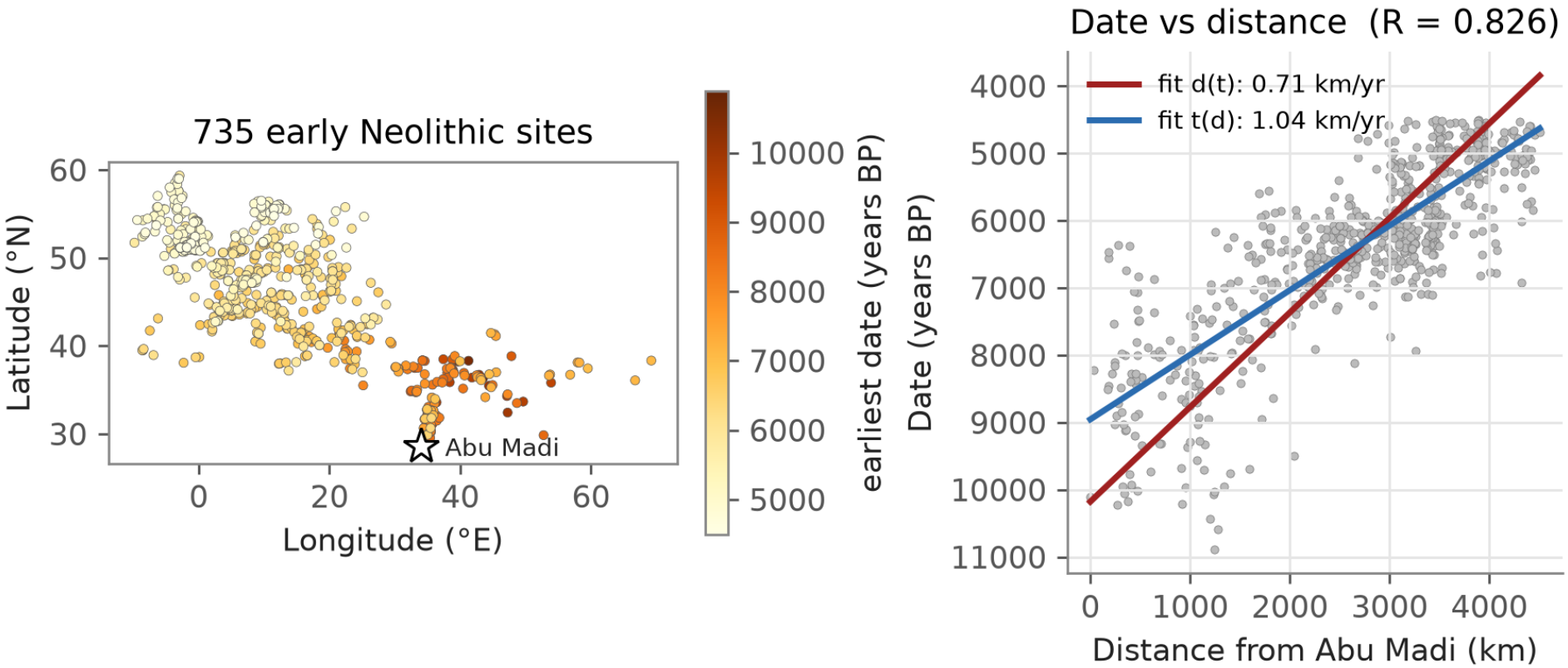
Farming spread from the Near East into Europe. How fast did the farming front move, and can a reaction–diffusion model predict this speed?

Data and method

- 735 early Neolithic sites in Europe, Anatolia and the Near East, each with its earliest radiocarbon date: from $10\,890$ to $4\,500$ years BP.
- Distance of every site from the probable origin, Abu Madi (Sinai).
- Linear regression of distance against date $\Rightarrow$ speed of the front.

5. Results and Discussion

Case Study: the Data



Our refit of the published data

$R = 0.826$; front speed 0.71–1.04 km/year from the two regressions (paper: $R = 0.827$, 0.71 ± 0.04 and 1.04 ± 0.05 km/year).

5. Results and Discussion

Case Study: Model and Parameters

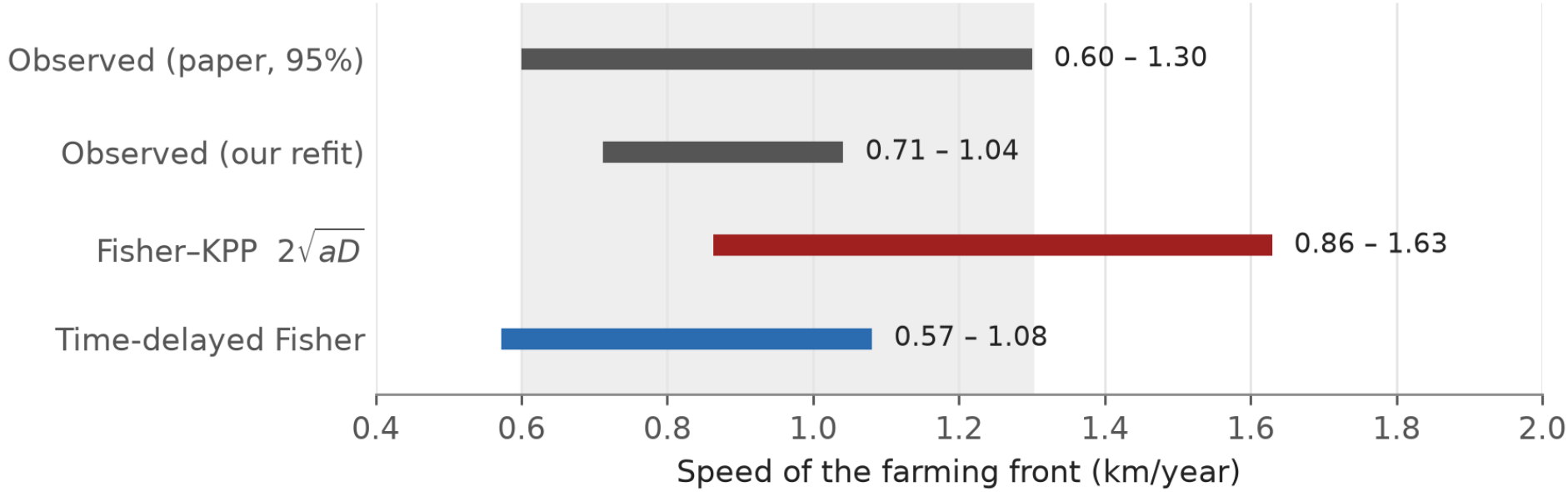
Fisher–KPP speed, without and with a generation time T [14], [15]

$$v_{\text{Fisher}} = 2\sqrt{aD}, \qquad v_{\text{delay}} = \frac{2\sqrt{aD}}{1 + aT/2}, \qquad D = \frac{m}{4T}$$

Parameter	Value (from the paper)	Meaning
a	0.029–0.035 /year	initial growth rate of farmers
m	900–2200 km^2	mobility (per generation)
T	29–35 years	generation time
$D = m/4T$	6.4–19.0 km^2/year	diffusion coefficient
v_{Fisher}	0.86–1.63 km/year	classic Fisher–KPP
v_{delay}	0.57–1.08 km/year	time-delayed Fisher

5. Results and Discussion

Case Study: Model vs Data

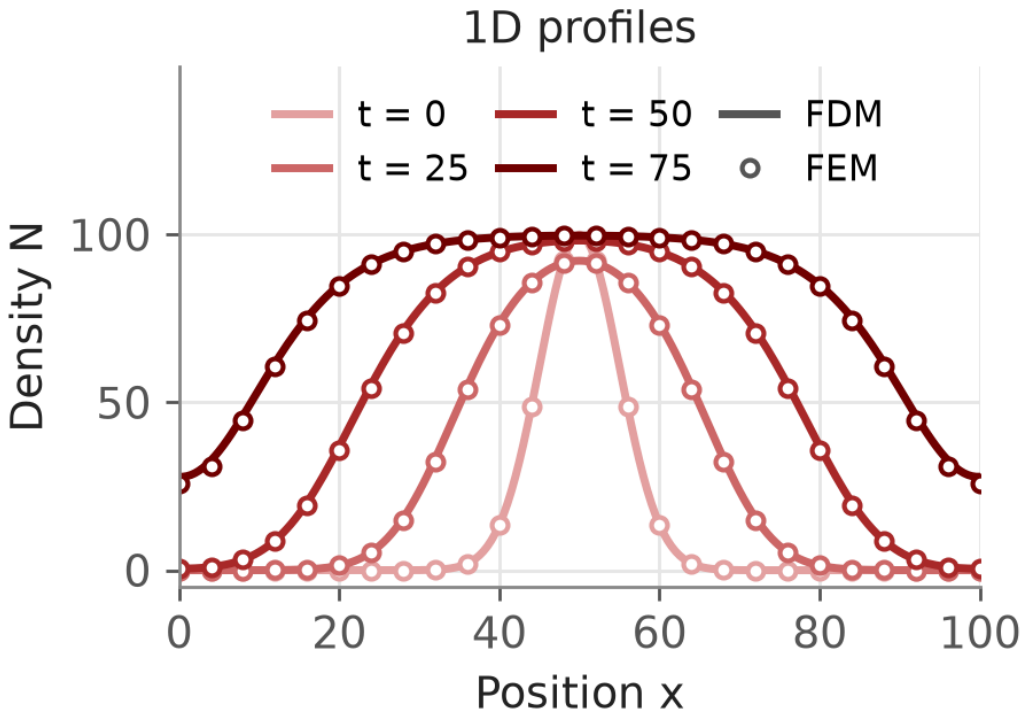


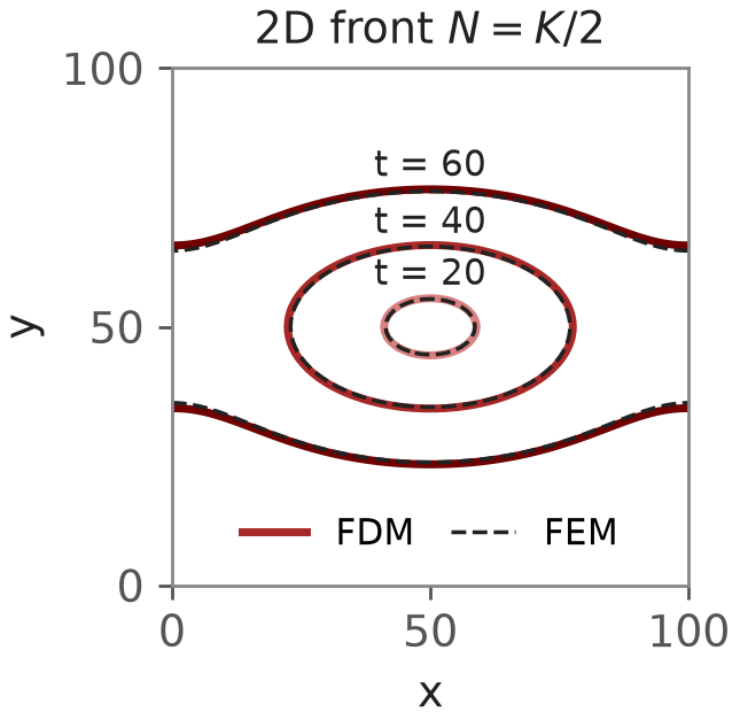
Result

The Fisher–KPP speed has the right order of magnitude (~ 1 km/year) but is partly too fast. With the generation time, the predicted 0.57–1.08 km/year lies inside the observed 0.6–1.3 km/year: farming spread with moving farmers, as a reaction–diffusion front.

5. Results and Discussion

Comparison of FDM and FEM



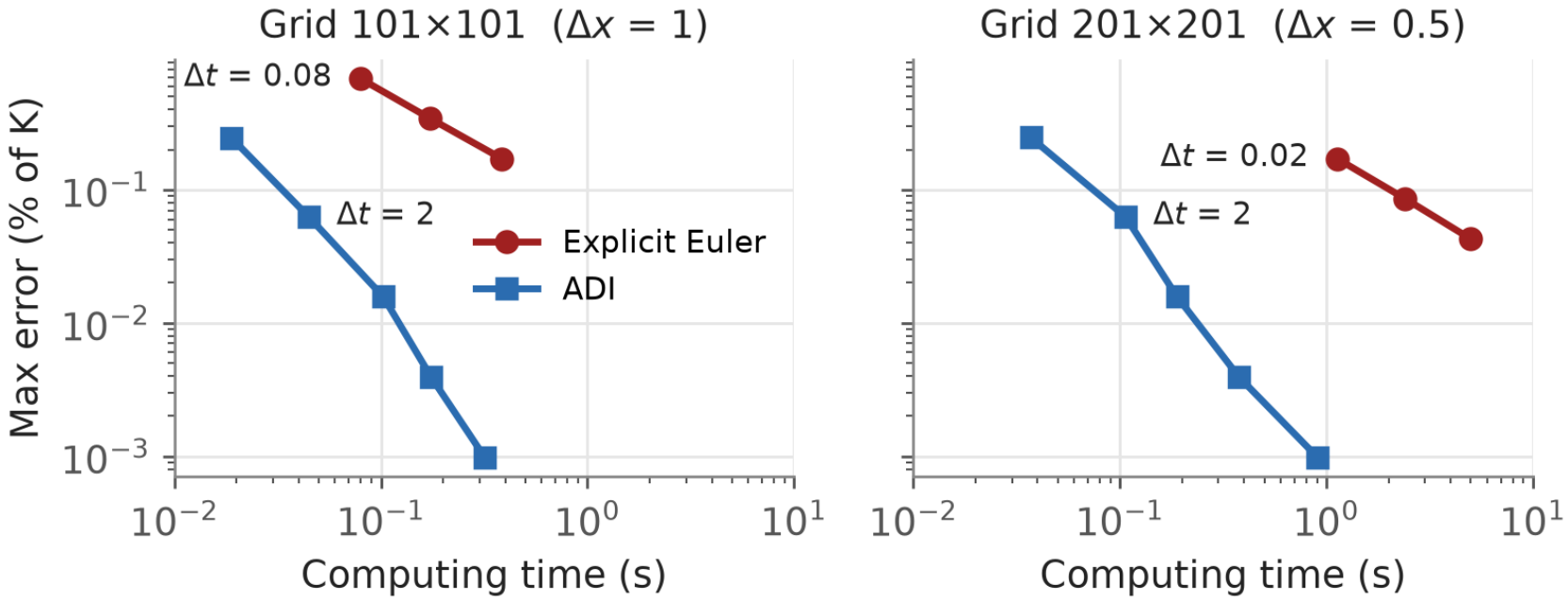


Result

With a time step 6–$10\times$ larger, FEM stays within 3% of K of the FDM solution (max difference: 2.2% in 1D, 2.6% in 2D).

5. Results and Discussion

Explicit Euler vs ADI (2D, $t = 60$)



Result

Grid 201×201: ADI ($\Delta t = 2$) is $\approx 10\times$ faster and $3\times$ more accurate than explicit Euler ($\Delta t = 0.02$).
Halving Δx: explicit Euler $\approx 13\times$ slower, ADI only $\approx 2.5\times$.

5. Results and Discussion

Discussion

What the results show

- **Homogeneous habitat:** the population forms a front that moves at an almost constant speed, close to $c = 2\sqrt{rD}$.
- **Real terrain (Ba Vì):** slopes slow the front ($\approx 0.35\,c$), wide rivers stop it, the poor habitat of the summit is never colonized.
- **Real data (Neolithic Europe):** the Fisher–KPP speed has the right order of magnitude; a generation time brings it inside the observed range.
- **Numerics:** FDM, FEM and ADI agree; FEM and ADI allow much larger time steps.

Limitations

- The Ba Vì parameters are illustrative, not fitted to a species.
- Deterministic model: no randomness, no long jumps, no bridges or roads.
- D, r, K do not change in time (no seasons, no climate change).

Conclusion

6. Conclusion

Conclusion

- We extended the Logistic Growth Model from an ODE to a Reaction-Diffusion PDE to describe spatial population dynamics.
- The model captures both population growth and spatial diffusion.
- Numerical simulations illustrate how the population spreads and approaches the carrying capacity over time.
- Different diffusion coefficients can produce different spatial spreading patterns.

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Thank You! Questions & Discussion

Summer Program on Mathematical Modeling 2026

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